

Amirreza Sokhankhosh

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HIGHLIGHT OF RELEVANT SKILLS

- **Data Science & Statistical Modeling:** Master of Science in Computer Science graduate with expertise in **causal inference**, **regression models**, **hypothesis testing**, **fairness analysis**, and **predictive modeling**, applied to real-world domains such as smart buildings and transportation.
- **Data Wrangling & Visualization:** Skilled in cleaning, transforming, and analyzing large datasets using **Python (Pandas, NumPy)** and **R**, with expertise in Matplotlib, Seaborn, and other visualization tools for EDA and reporting. Experienced in building data dashboards using **React** and **Tableau**.
- **Machine Learning & Forecasting:** Hands-on experience in time series forecasting, classification, NLP, and LLM-powered applications using Scikit-learn, PyTorch, TensorFlow, and Hugging Face Transformers.
- **Databases, Geospatial & Data Systems:** Proficient in **SQL** and **NoSQL** databases (PostgreSQL, MongoDB, MySQL), including schema design, **efficient querying**, integrating analytics pipelines with production data systems, and working with geographical/spatial data in **PostGIS** and **Tableau**.
- **Collaboration & Applied Research:** Experienced in academic and industry settings, applying **Agile methodologies**, contributing to **cross-functional teams**, and communicating **data-driven insights** to stakeholders.

EXPERIENCE

University of Manitoba

ML Engineer (Contract)

May 2025 – Present

Winnipeg, Canada

- Preprocessed and normalized large **IoT datasets** in, producing model-ready inputs that stabilized convergence.
- Built a scalable ML pipeline with **PyTorch**, enabling secure, **parallel training** across 8+ distributed nodes with zero raw data exposure.
- Designed and deployed a Temporal Fusion Transformer for **multi-variate time-series forecasting**, reducing prediction errors by >20% for temperature, CO₂, and humidity data.
- Engineered a blockchain-integrated backend using Hyperledger Fabric and Express.js to automate model aggregation for 76 IoT sensor streams.

University of Manitoba

Graduate Research Assistant

Sep 2023 – Jul 2025

Winnipeg, Canada

- Designed **4** end-to-end distributed **ML pipelines** with **PyTorch** and **TensorFlow**, using modular code and containerization, supporting up to **36** training nodes.
- Built a microservice-based orchestration layer with **Flask** and **Express.js** and a fast **REST API** to coordinate distributed training reliably, achieving **90–100%** node uptime across all runs.
- Improved **model robustness** via adaptive validation schemes, handling inconsistent or adversarial updates, achieving **100%** detection of simulated adversaries.
- Optimized **training efficiency** by offloading heavy computation to servers, enabling lightweight clients, and reducing iteration time from **hours to minutes**.
- Co-authored **5+** **research papers** in distributed ML engineering, published/submitted to **top IEEE venues**.

Bobo App Ltd.

Full Stack Developer Intern

May 2024 – Aug 2024

Winnipeg, Canada

- Collaborated with a team at **MILA** in developing an efficient **chatbot** responsible for **customer service**.
- Accelerated product development by designing and implementing **RESTful APIs** using Supabase and **PostgreSQL**, enhancing backend functionality and data management.
- Automated data integration by developing a Python script to convert CSV data into executable SQL, significantly streamlining **database population**.

- Ensured seamless **full-stack integration** through close collaboration with the front-end team to align API specifications with user interface requirements.
- Contributed to **agile** workflow efficiency by utilizing the Atlassian suite (**Jira, Confluence**) for task management, documentation, and team coordination.

K.N. Toosi University of Technology

Jun 2021 – Aug 2022

Research Assistant

- Led an undergraduate research team to investigate the fairness of solar panel funding in Switzerland, culminating in key analytical findings.
- Executed comprehensive **data collection and cleaning** of over **10GB of raw data** from diverse government portals, ensuring high data integrity for advanced analysis.
- Leveraged advanced **causal inference** techniques in **R** to analyze complex datasets, producing detailed analytical reports that quantified and highlighted significant subsidy biases.

Nadin Soft (Sadr Group Company)

Jul 2020 – Dec 2020

Full-stack Developer Intern

- Drove the successful completion of a critical internal project, addressing key deficiencies in **API security, authentication, and data modeling** for production deployment.
- Developed robust RESTful APIs with AdonisJS and **Express.js**, implementing **JWT authentication** to secure user access and enable core application functionalities.
- Designed and implemented highly accurate MySQL database schemas for precise tracking and management of employee working and non-working hours, foundational for the project's business logic.
- **Deployed** fully functional features to the **live production server** and created thorough Swagger API documentation, facilitating seamless internal integration and knowledge transfer.
- Contributed to an Agile Scrum development team by actively tracking and managing personal tasks within ClickUp, ensuring efficient progress and clear communication within sprints and **CI/CD pipelines**.

NYC Taxi Analytics Dashboard

Jan 2025 – Mar 2025

Personal Project | [Code](#)

- Built an end-to-end **data pipeline** in Python with **BigQuery**, aggregating NYC taxi trips into zone-level metrics and reducing local processing overhead to near-zero.
- Developed a **React dashboard** with **Material-UI, Plotly, and Leaflet**, visualizing 263 taxi zones via interactive heatmaps, time-series views, and Sankey flows.
- Engineered **geospatial processing** using GeoPandas, shapefiles, and coordinate transformations.
- Optimized **data performance** by offloading heavy aggregations to BigQuery, using caching and client-side CSV parsing, achieving smooth rendering across devices.

EDUCATION

University of Manitoba

Sep 2023 – Aug 2025

Master of Science in Computer Science (GPA: 4.4 / 4.5)

Winnipeg, Canada

- **Relevant Coursework:** Security & Privacy, Deep Generative Modeling, Data Mining: A+

K.N. Toosi University of Technology

Sep 2018 – Feb 2023

Bachelor of Science in Computer Engineering

TECHNICAL SKILLS

Data Science & Analytics: Python (Pandas, NumPy, SciPy, Scikit-learn), R, SQL, Matplotlib, Seaborn, Plotly.
Machine Learning & Forecasting: Time Series (PyTorch Forecasting), NLP (Transformers), TensorFlow, PyTorch.
Databases & Data Systems: PostgreSQL, MySQL, MongoDB, Supabase, Redis.
Programming Languages: Python, R, SQL, Java, JavaScript, TypeScript, Node.js.
Cloud & DevOps: AWS, Docker, Git, GitHub Actions, CI/CD, Linux.
Web Frameworks: *Back-end:* Flask, Django, Express.js, NestJS. *Front-end:* React, Tailwind.
Tools & Methodologies: Jira, Confluence, Swagger, Agile, Scrum.